Biochemistry Cume
Enzymes Total points: **100 points**

To receive credit, **A**. Answers should be clearly explained. Ambiguous or partially correct answers will not receive any credit. **B**. All steps involved in arriving at a solution to a problem should be provided.

1. **A.** What assumption concerning substrate concentration is made in the discussion of enzyme kinetics? Why is it important? (**5 points**)

B. Write the Briggs-Haldane equation that characterizes steady state enzyme kinetics. (**5 points**)

C. Write the Michealis-Menten equation. (**5 points**)

D. Define, **a.** K_M and **b.** k_{cat} **10 (5 + 5) points**

F. List the techniques that are commonly used to study enzyme-transition state complementarity. (**8 points**)

G. Why is kcat/K_M, the specificity constant, the most useful parameter for discussing catalytic efficiency of an enzyme? (**8 points**)

H. Explain why is a regulatory enzyme very often the first enzyme in a multiple enzyme sequence? (**4 points**)

I. What are ping-pong reactions? Give one example of a ping-pong reaction. (**10 points**)

J. At what substrate concentration will an enzyme with a k_{cat} of 30 sec^{-1} and a K_M of 0.005 M show 75% of its maximum rate? (**20 points**)

K. Carbonic anyhydrase from erythrocytes (Molecular mass = 30 kDa) has one of the highest turnover numbers among known enzymes. It catalyzes the reversible hydration of CO_2,

$$H_2O + CO_2 \leftrightarrow H_2CO_3$$

If 10 μg of pure carbonic anyhydrase catalyzes the hydration of 0.30 g of CO2 in 1 minute at 37 °C at V_{max}, what is the turnover number of carbonic anhydrase (in units of $minute^{-1}$) (**25 points**)